

A short parallel-query proof for fourfold forrelation

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Abstract

A single parallel quantum interrogation of four binary phase masks needs hard photon dose at least $(2/15)N^{1/8}$ to decide fourfold forrelation with error at most $1/3$, for powers of two $N \geq 2^{30}$. The proof has three parts: a matrix bound on Fourier coefficients, a correlated sign distribution, and concentration onto the exact promise. Arbitrary idlers, repeated modes, and coherence between photon-number sectors are allowed. Outcome-dependent fresh batches require an additional Fourier bound.

1 Statement and proof outline

Let $N = 2^d$, with coordinates in $\mathbb{F}_2^d$, and set

$$H_{ij} = N^{-1/2}(-1)^{i \cdot j}, \quad u = N^{-1/2}(1, \dots, 1)^\top.$$

For four sign vectors $x^{(b)} \in \{\pm 1\}^N$ and $D_b = \text{diag}(x^{(b)})$, define

$$F(x) = \text{Forr}_4(x) = u^\top D_1 H D_2 H D_3 H D_4 u.$$

The task is to distinguish $F \geq 1/4$ from $F \leq -1/4$. A parallel probe is prepared before any sample output is available, passes through the direct-sum mask $D_1 \oplus \dots \oplus D_4$ once, and is measured arbitrarily. Hard dose D bounds the total number of signal photons on the entire support of the prepared state. References and idlers bypass the masks and are free.

Theorem 1. *For every power of two $N \geq 2^{30}$, a parallel probe solving the task with worst-case error at most $1/3$ has*

$$D \geq \frac{2}{15}N^{1/8}.$$

This includes mixtures of probes and predetermined collections of batches whose total hard dose is D : they can be prepared jointly and measured by a single final POVM.

We construct input laws μ_+, μ_- with mean forrelation approximately $\pm(2\rho/\pi)^3$, where $\rho = 200/201$. Their low-order sign correlations are small. Lemma 2 limits how strongly a parallel measurement can collect those correlations. Concentration then puts the mass of each law predominantly on its required promise.

The proof below uses standard Gaussian Hermite expansion, the Gaussian arcsine identity, and the Efron–Stein and Gaussian Poincare inequalities. The arithmetic check `verify_constants.py` verifies the final constants. See `./AUDIT.md` for the proof audit.

2 The parallel Fourier bound

Write $m = 4N$ for the total number of sign coordinates, and $\chi_S(x) = \prod_{a \in S} x_a$. A bosonic probe of hard dose D embeds in D labelled slots with alphabet $\{0, 1, \dots, m\}$, where 0 denotes vacuum. Send each normalized occupation basis vector with $n \leq D$ photons to the normalized symmetric sum of distinct slot words with those labels and $D - n$ vacuum symbols. These sums are orthonormal, giving an isometry from the truncated Fock space. The query is exactly $O_x^{\otimes D}$, with $O_x|0\rangle = |0\rangle$ and $O_x|a\rangle = x_a|a\rangle$. Enlarging to arbitrary slot states can only strengthen the competitor.

For a word $a = (a_1, \dots, a_D)$ let ∂a be the set of nonzero labels occurring oddly. Purify the input and write $\psi = \sum_a |a\rangle \otimes \psi_a$, where the ψ_a are idler vectors. For an acceptance effect $0 \leq E \leq I$, put $E_{ab} = (\langle a| \otimes I)E(|b\rangle \otimes I)$. Then

$$f(x) = \sum_{a,b} \chi_{\partial a \triangle \partial b}(x) \langle \psi_a, E_{ab} \psi_b \rangle, \quad \sum_a \|\psi_a\|^2 = 1. \quad (1)$$

Thus $\widehat{f}(S) = 0$ for $|S| > 2D$. Arbitrary idlers are included in the operator blocks E_{ab} . The Fourier coefficient is $\widehat{f}(S) = 2^{-m} \sum_x f(x) \chi_S(x)$.

We use the elementary Schur factorization bound. If $K_{ab} = \langle p_a, q_b \rangle$ with $\|p_a\| \leq A$ and $\|q_b\| \leq B$, then

$$\|K_{ab} E_{ab}\| \leq AB \|E\|. \quad (2)$$

Map $(v_b)_b$ to $(q_b \otimes v_b)_b$, apply $I \otimes E$, and contract with the p_a to obtain (2). Let $\gamma_2(K)$ be the infimum of AB .

Lemma 2. For arbitrary complex numbers c_S at level $1 \leq r \leq 2D$,

$$\begin{aligned} \left| \sum_{|S|=r} c_S \widehat{f}(S) \right| &\leq \frac{1}{2} \max_{|S|=r} |c_S| \sum_{k=0}^r \binom{D}{k} \binom{D}{r-k} m^{\min(k, r-k)/2} \\ &\leq \frac{1}{2} \binom{2D}{r} m^{r/4} \max_{|S|=r} |c_S|. \end{aligned} \quad (3)$$

Proof. In a pair of words (a, b) with $|\partial a \triangle \partial b| = r$, each differing label occurs oddly on exactly one side. Mark its last occurrence on that side. This gives a unique record of r marked positions among the D ket and D bra slots.

Fix a record with k ket marks. The marked labels are tuples $\alpha(a)$ of length k and $\beta(b)$ of length $r - k$. Let A, B be their underlying sets and put

$$S_a = \partial a \triangle A, \quad S_b = \partial b \triangle B.$$

A word is admissible on its own side if its marked labels are nonzero and distinct, each mark is the last occurrence of its label, and its residual parity is disjoint from its marked set. Write these two indicators as $\ell(a)$ and $r(b)$. Define

$$C_{\alpha\beta} = \begin{cases} c_{A \cup B}, & \text{all } r \text{ marked labels are distinct,} \\ 0, & \text{otherwise.} \end{cases}$$

The exact record contribution to (1) is the Schur symbol

$$K_{ab}^R = \ell(a) r(b) C_{\alpha(a), \beta(b)} \mathbf{1}_{\{S_a = S_b\}}. \quad (4)$$

Residual equality cancels every unmarked parity. The two disjointness conditions put each marked label on its specified odd side, and the last-occurrence conditions make the record unique. Repeated labels and vacuum slots are retained.

A factorization of C lifts to (4) by tensoring its row and column factors with $|S_a\rangle$ and $|S_b\rangle$, then multiplying by the indicators. Thus $\gamma_2(K^R) \leq \gamma_2(C)$. The matrix C has at most m^k rows and m^{r-k} columns. Standard basis vectors on its shorter side give

$$\gamma_2(C) \leq m^{\min(k, r-k)/2} \max |c_S|.$$

Since $r > 0$, K^R has zero diagonal: at $a = b$, residual equality forces $A = B$, while distinctness forces $A \cap B = \emptyset$, contradicting $|A| + |B| = r$. Hence

$$[K_{ab}^R E_{ab}] = [K_{ab}^R (E - I/2)_{ab}], \quad \|E - I/2\| \leq 1/2.$$

Apply (2), and sum over the $\binom{D}{k} \binom{D}{r-k}$ records. Vandermonde's identity finishes the proof. $\square$

3 Correlated signs with small moments

For $s = 1, 2, 3$, independently draw standard Gaussian vectors Z_s, W_s with cross-covariance $\mathbb{E} Z_s W_s^\top = \rho H$. This is a valid covariance because H is orthogonal and $\rho < 1$. Set $a_s = \text{sgn } Z_s$, $b_s = \text{sgn } W_s$, applying sgn componentwise, and form

$$y = (a_1, b_1 \odot a_2, b_2 \odot a_3, b_3).$$

Here $\odot$ denotes componentwise multiplication. Choose three independent uniform signs $\zeta_1, \zeta_2, \zeta_3$, put $\zeta_4 = \sigma \zeta_1 \zeta_2 \zeta_3$, and set $x^{(b)} = \zeta_b y^{(b)}$. This defines μ_σ .

For one link let $L(A, B) = \mathbb{E} \prod_{i \in A} a_i \prod_{j \in B} b_j$. If $a = |A|$, $b = |B|$ and $a + b \leq r \leq \sqrt{N}$, then

$$|L(A, B)| \leq \left(\frac{r}{2\sqrt{N}} \right)^{\max(a, b)}. \quad (5)$$

If one set is empty the moment is zero unless both are empty. Otherwise the cross-covariance matrix $R = \rho H_{A, B}$ has norm at most $\rho \sqrt{ab/N} \leq r/(2\sqrt{N})$. The sign products have Hermite degrees at least a and b . Conditional expectation pairs only equal degrees $p \geq \max(a, b)$, where it acts as $R^{\otimes p}$ on symmetric tensors and has norm at most $\|R\|^p$. Cauchy–Schwarz and the unit L^2 norms give (5).

For $S = (S_1, \dots, S_4)$ of total size r , gauge averaging kills the difference of its two moments unless every $q_b = |S_b|$ is odd. In that case

$$(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-}) \chi_S = 2L(S_1, S_2)L(S_2, S_3)L(S_3, S_4).$$

Taking weights $(3/4, 1/4)$, $(1/2, 1/2)$ and $(1/4, 3/4)$ in the three maxima proves

$$\max(q_1, q_2) + \max(q_2, q_3) + \max(q_3, q_4) \geq 3r/4.$$

Consequently

$$|(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-}) \chi_S| \leq 2^{1-3r/4} r^{3r/4} N^{-3r/8}. \quad (6)$$

Only even levels $r \geq 4$ can contribute.

4 Concentration onto the promises

The arcsine identity gives $\mathbb{E}a_i b_j = \beta_N H_{ij}$, with

$$\beta_N = \frac{2\sqrt{N}}{\pi} \arcsin(\rho/\sqrt{N}).$$

The three links are independent. The defining $N^{-5/2}$ path-sum coefficient, three factors $N^{-1/2}$, and N^4 terms give

$$\mathbb{E}_{\mu_\sigma} F = \sigma \beta_N^3, \quad \beta_N^3 - \frac{1}{4} > \left(\frac{1400}{2211}\right)^3 - \frac{1}{4} > \frac{1}{260}. \quad (7)$$

Here $\arcsin z \geq z$ and $\pi < 22/7$ suffice.

Lemma 3. *For either sign law, $\text{Var } F < 1540/N$.*

Proof. Represent each link with independent standard Gaussians G, ξ, η :

$$a = \text{sgn}(G + \xi/\sqrt{200}), \quad b = \text{sgn}(HG + \eta/\sqrt{200}).$$

After standardization its correlation is ρH . Conditional on G_1, G_2, G_3 , all coordinates of the four ungauged blocks y are independent. Primitive conditional means are $\phi(G)$ and $\phi(HG)$, where

$$\phi(z) = \text{erf}(10z), \quad L = \|\phi'\|_\infty = 20/\sqrt{\pi}, \quad L^2 < 128.$$

The function ϕ is applied componentwise to vectors. The gauges multiply F by the deterministic sign σ .

For an independent biased sign Y , Efron–Stein bounds its variance contribution by $(1 - (\mathbb{E}Y)^2)(\partial_Y F)^2 \leq (\partial_Y F)^2$, where $\partial_Y F$ is the coefficient of Y . Each endpoint block has total squared derivative $1/N$. For block two the derivative is $\ell_j r_j$, with

$$\ell = HD_{a_1} u, \quad r = HD_{b_2 \odot a_3} HD_{b_3} u.$$

These vectors are independent, $\mathbb{E}\ell_j^2 = 1/N$, and $\|r\| = 1$. The contribution is $1/N$ after unconditional averaging. Block three is the same calculation with the ends reversed. Hence

$$\mathbb{E} \text{Var}(F \mid G_1, G_2, G_3) \leq 4/N. \quad (8)$$

Put $M(g) = D_{\phi(g)} HD_{\phi(Hg)}$, a contraction. The conditional mean is

$$\bar{F} = \sigma u^\top M(G_1) M(G_2) M(G_3) u.$$

At one link write this as $p^\top M(G) q$, up to σ . The partial-chain vectors p, q are independent of each other and G , with norms at most one and $\mathbb{E}|p_i|^2, \mathbb{E}|q_i|^2 \leq 1/N$. To justify the last statement, set $(P_s v)_i = v_{i+s}$ and $(D_s v)_i = (-1)^{s \cdot i} v_i$. The Sylvester identities and oddness of ϕ give

$$M(P_s g) = P_s M(g), \quad M(D_s g) = M(g) P_s.$$

Replacing the leftmost Gaussian by $P_s G$ in a right chain, or by $D_s G$ in a left chain written with transposed factors, translates the vector by P_s . Gaussian invariance and transitivity give equal coordinate second moments, whose sum is at most one; the empty chain is u .

Differentiation, using $H = H^\top$, gives

$$\|\nabla_G \bar{F}\|^2 \leq 2L^2 (\|p \odot HD_{\phi(HG)} q\|^2 + \|q \odot HD_{\phi(G)} p\|^2).$$

Independence and the second-moment bounds make each expected squared norm at most $1/N$. Gaussian Poincare over the three links gives $\text{Var } \bar{F} \leq 12L^2/N$. With (8), $\text{Var } F \leq (4 + 12L^2)/N < 1540/N$. $\square$

5 Closing the bound

By (7) and Chebyshev, for $N \geq 2^{30}$,

$$\varepsilon_\sigma := \mu_\sigma\{\sigma F < 1/4\} \leq 104104000/N < 1/10. \quad (9)$$

Pointwise correctness of an acceptance probability $f \in [0, 1]$ gives

$$\mathbb{E}_{\mu_+} f \geq \frac{2}{3}(1 - \varepsilon_+), \quad \mathbb{E}_{\mu_-} f \leq \frac{1}{3}(1 - \varepsilon_-) + \varepsilon_-.$$

Consequently

$$\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f \geq \frac{1}{3} - \frac{2}{3}(\varepsilon_+ + \varepsilon_-) > \frac{1}{5}. \quad (10)$$

Suppose $D < (2/15)N^{1/8}$. Then $2D < \sqrt{N}$. Combine (3) and (6), recall $m = 4N$, and use $\binom{2D}{r} \leq (2eD/r)^r$:

$$\begin{aligned} |\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f| &\leq \sum_{\substack{4 \leq r \leq 2D \\ r \text{ even}}} \left(2^{3/4} e D r^{-1/4} N^{-1/8}\right)^r \\ &\leq \frac{q^4}{1 - q^2} < \frac{1}{12}, \quad q = 2^{1/4} e D N^{-1/8} < \frac{1}{2}. \end{aligned}$$

Here $r^{-1/4} \leq 4^{-1/4}$, $e < 3$, and $2^{1/4} < 5/4$. Since $1/12 < 1/5$, this contradicts (10) and proves Theorem 1.

6 What remains for adaptive batches

For an outcome-dependent fresh protocol, acceptance has degree at most $2D$: multiply batch probabilities along each classical history. A sufficient remaining estimate is

$$\left| \sum_{|S|=r} c_S \hat{f}(S) \right| \leq \binom{2D}{r} (4N)^{r/4} \max_{|S|=r} |c_S| \quad (1 \leq r \leq 2D). \quad (11)$$

With (11), the same even-level calculation gives a gap at most $2q^4/(1 - q^2) < 1/6 < 1/5$, proving the same dose bound. This estimate remains unproved for adaptive batches; `../AUDIT.md` describes the gap in the earlier argument. The theorem established here concerns parallel interrogation.